Announcements

- Any questions/issues?
- Schedule:

  Today (Follow-up): Question 2 from "Forcing and Timescales" discussion — Exponential Growth and Doubling
  Today (New): Modeling and forecast of climate change
  Tomorrow (Lab): Computational Modeling (web app)
  Thursday (Discussion): More Computational Modeling
  Next Week: Oceans and Ice

Today's Reading Quiz (3 min)

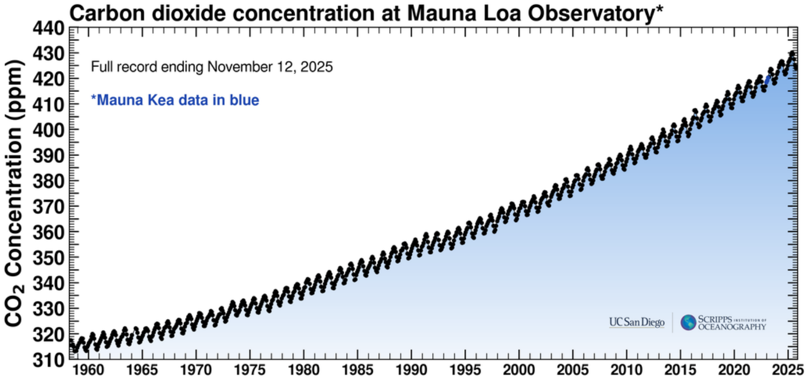
No reading for today, but please prepare for tomorrow and Thursday by:

- Reading Chapter 8 from Dessler ("Predictions of Future Climate Change")
- Watching the tutorial video for the climate simulator **EN-ROADS** (find link on "Weekly Reading Assignments" page)

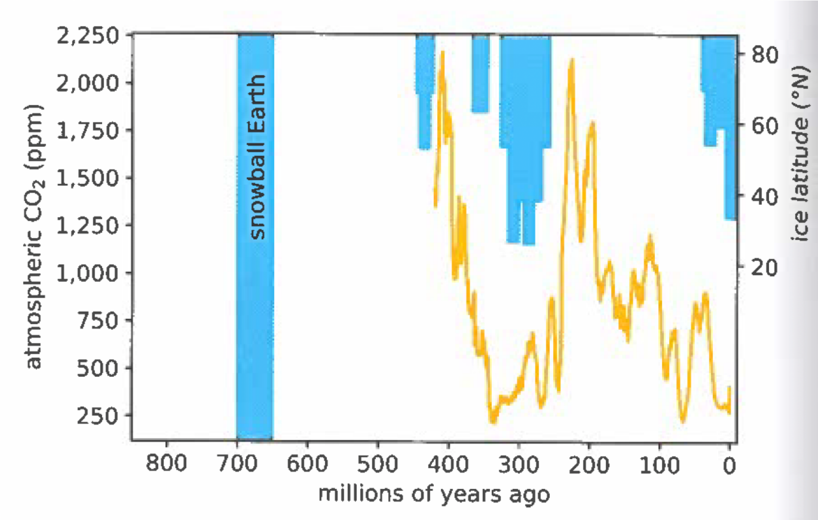
Final Poster Project

- Choose a topic that is interesting to you and make a poster to present (5min) to the class during our final exam period (Thurs, Dec 11, 10am).
- Work on your own or with a partner.
- Think of it as a Discussion/Lab/Lecture that you create, explaining some topic to yourself and us.
- **Talk to me by Thanksgiving (Tues Nov 25)** to verify your project topic (**worth one Discussion assignment**).
- In the last week of class you will have time (lecture and discussion) to work on your final poster project.

Exponential Growth of CO_2 concentration

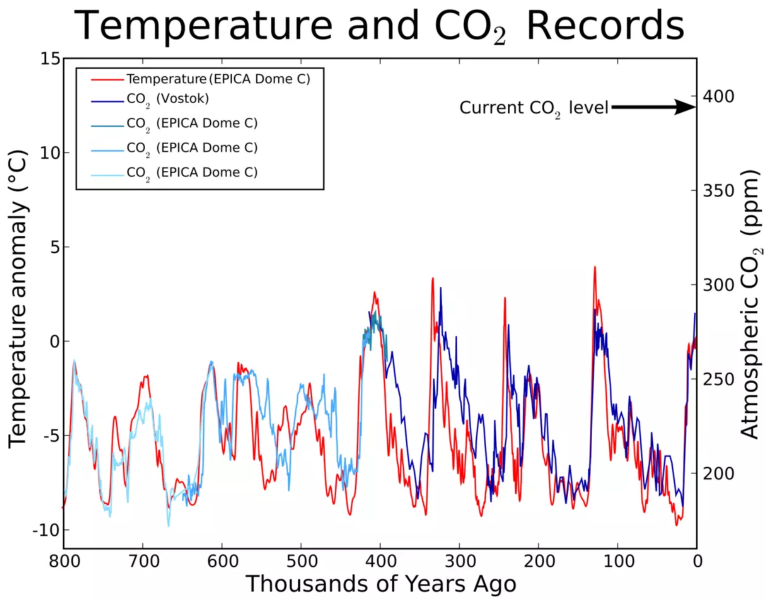


CO_2 concentration on planetary timescales



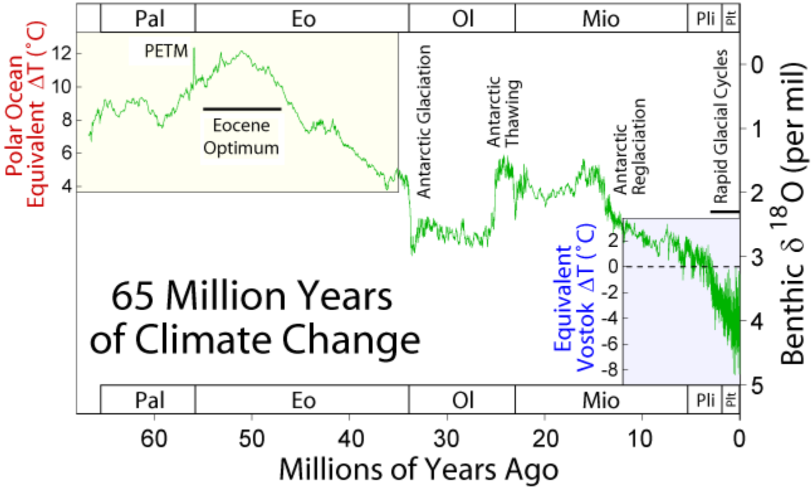
Dessler Fig 7.7

CO_2 and temperature



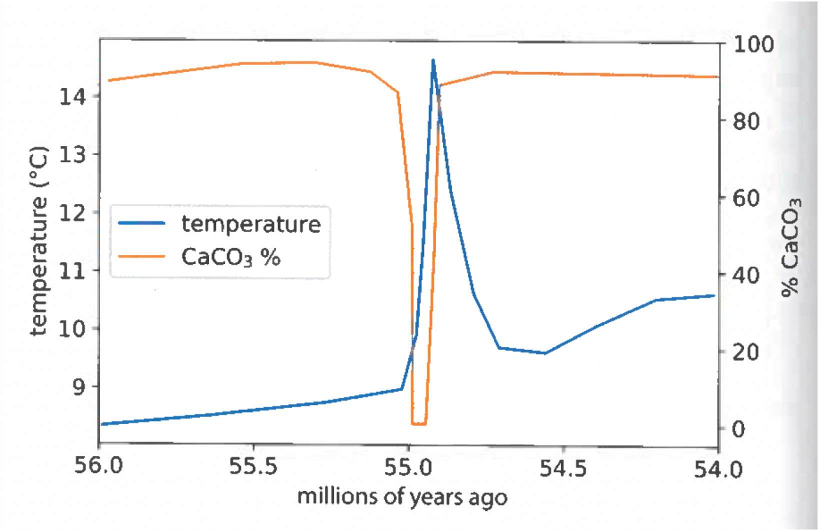
Carleton College SERC

The last big temperature/CO_2 surge: PETM



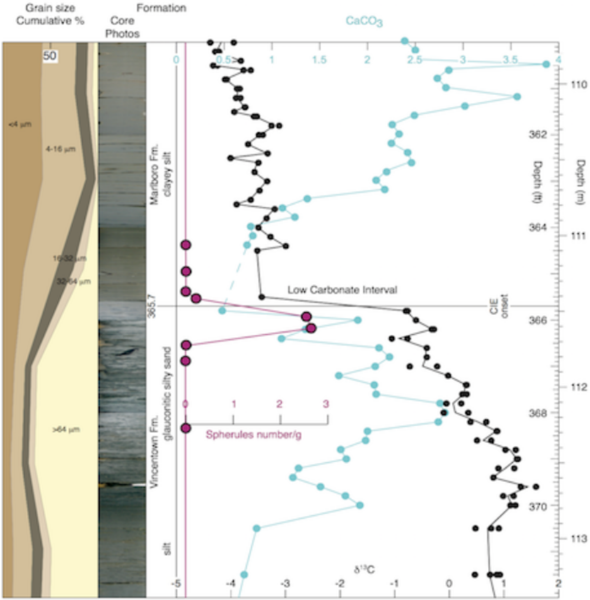
Wikimedia (Rohde)

PETM timescale



Dessler Fig 7.8

PETM timescale



Rutgers, Marlboro Clay Project

Timescales of contemporary climate change

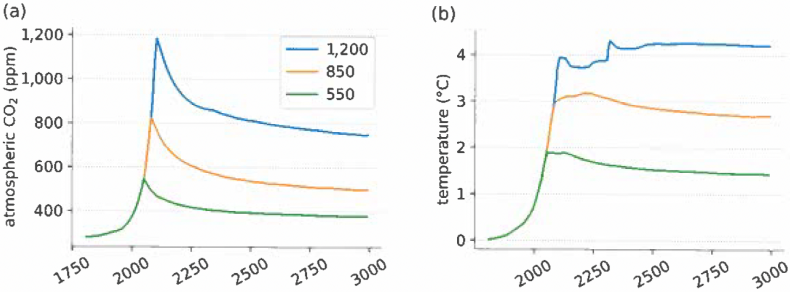


Figure 8.5 (a) Amount of carbon dioxide in the atmosphere as a function of time, for the next 1,000 years. Carbon dioxide emissions rise at 2 percent/year until it hits a peak abundance (550, 850, and 1,200 ppm); then emissions are decreased instantly to zero. (b) The temperature time series corresponding to each carbon dioxide time series. Adapted from Figure 1 of Solomon et al. (2009).

Dessler